

ADVANCED COMPUTATIONAL BIOCHEMISTRY HACKATHON

Sapienza Università di Roma | Masters in Computational Biology & Biochemistry

Activity 1: Clinical Variant Prioritization

Diagnostics & Precision Medicine

The Scenario: A biotech diagnostics company needs rapid computational assessment of newly identified patient mutations. Your team acts as **Computational Structural Biology Consultants**.

Learning Objectives

- Retrieve clinically relevant variants from NCBI ClinVar/dbSNP.
- Map mutations onto predicted AlphaFold protein structures.
- Evaluate biophysical and functional consequences.
- Prioritize variants (Pathogenic, Benign, or VUS).

Suggested Proteins

Protein	Disease Context
TP53	Somatic & Germline Cancer
CFTR	Cystic Fibrosis
PAH	Phenylketonuria (PKU)
BRCA1	Hereditary Breast Cancer
G6PD	Enzymopathy

Activity 2: Evolutionary Structural Bioinformatics

Conservation vs. Divergence

The Scenario: "Can structure remain conserved while sequence evolves?" Investigate how evolutionary adaptation shapes the fold-space of homologous proteins across diverse taxa.

Learning Objectives

- Compare homologous proteins across diverse taxa.
- Analyze structural conservation despite sequence divergence.
- Infer evolutionary pressures (dN/dS, Co-evolution).
- Integrate sequence + structure evidence for phylogeny.

Target Families

Family	Evolutionary Context
Cytochrome c	Ultra-high conservation
HSP family	Stress & thermo-adaptation
DNA Pol	Functional fidelity constraints
LDH / CA	Metabolic/Convergent adaptation

The Scenario: A startup is exploring novel antimicrobial or cancer targets using AI-predicted structures. Your group functions as a **Computational Drug Discovery Team**.

Learning Objectives

- Identify candidate druggable regions (pockets/ allosteric).
- Integrate structure/function relationship analysis.
- Evaluate target feasibility (druggability scores).
- Propose computational screening strategies (Docking/MD).

Exploration Targets

Category	Candidate Targets
Microbial	Beta-lactamases, Gyrase, SARS-CoV-2
Human	Kinases, Mutant p53, PARP family

Technical Protocol & Deliverables

Software Toolkit

- **Structure:** AlphaFold DB, PyMOL, ChimeraX.
- **Analysis:** FoldX, Rosetta, or PopMusic.
- **Sequence:** BLAST, Clustal Omega, Jalview.

Final Submission

- A 10-slide technical presentation.
- Structural overlays highlighting key residues.
- Ranking table of variants or druggable sites.